2026 년도 KMRTS 추계학술대회

피드백 기반 인식 정제를 통한 열악한 해상 환경에서의 강인한 카메라–레이다 융합

PRFusion · Feedback-Driven Perception Refinement for Robust Maritime Camera–Radar Fusion



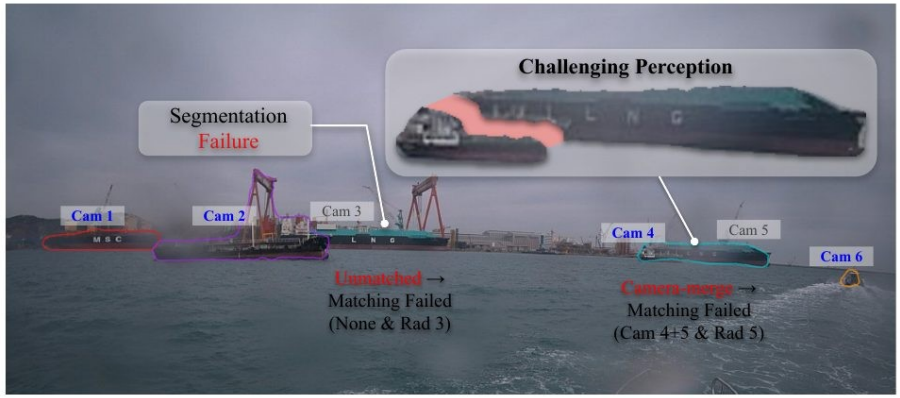
신영하 , 김진환 *

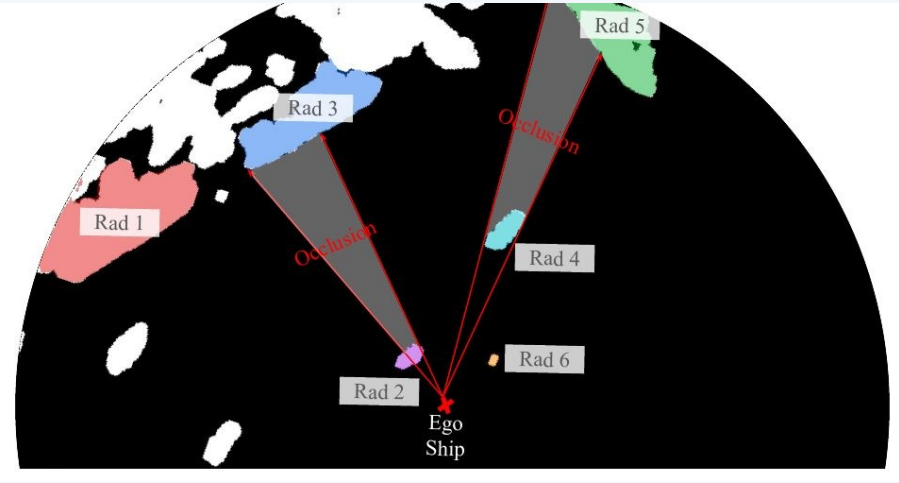
KAIST

* 교신 저자

PROBLEM

Association failures originate in segmentation





Camera-merge

Adjacent objects under occlusion collapse into one camera mask.

Radar-merge

Objects along the same bearing merge into one radar return.

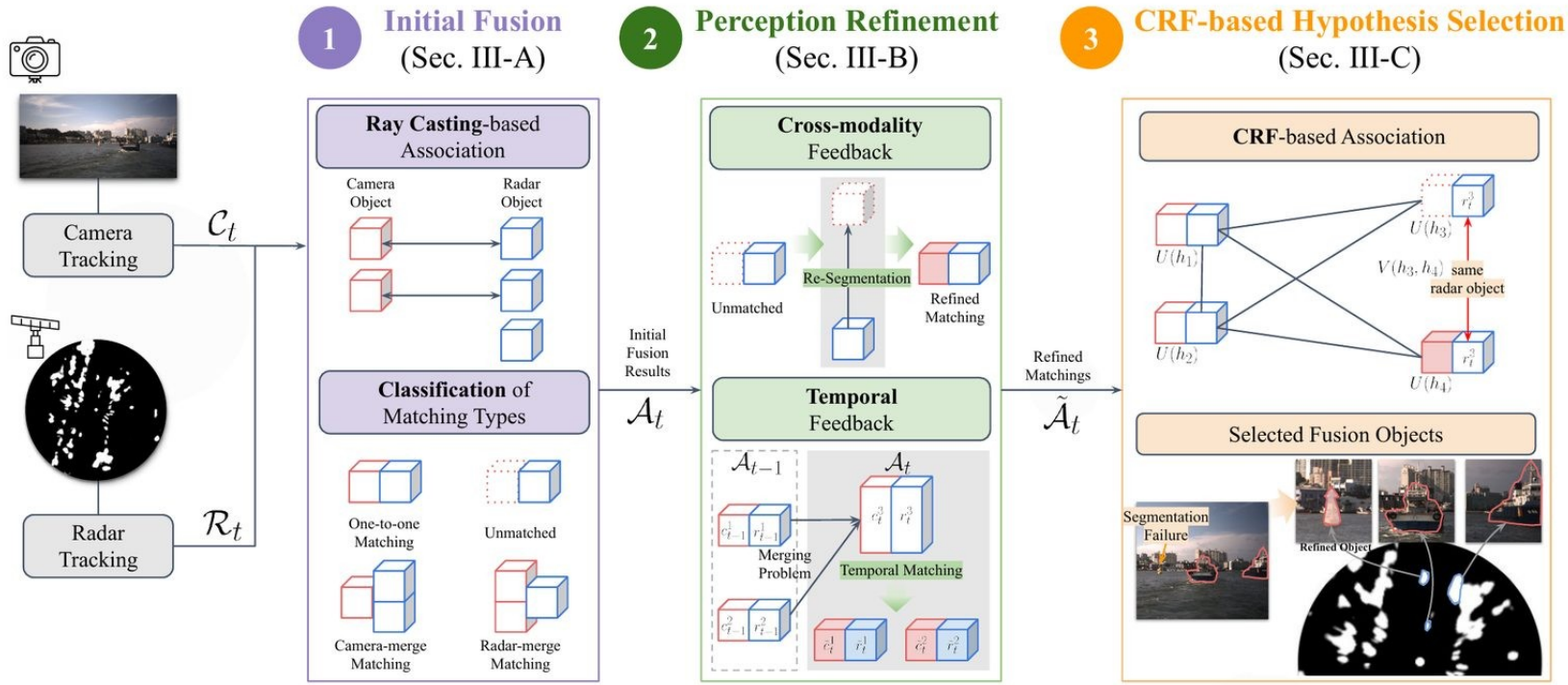
Unmatched

An object is detected by only one of the two sensors.

KEY IDEA **A failed association is not the final result, but a trigger to re-segment.**

PROPOSED METHOD

PRFusion framework overview



1 Initial fusion

Ray-casting association, classified into four matching types

2 Perception refinement

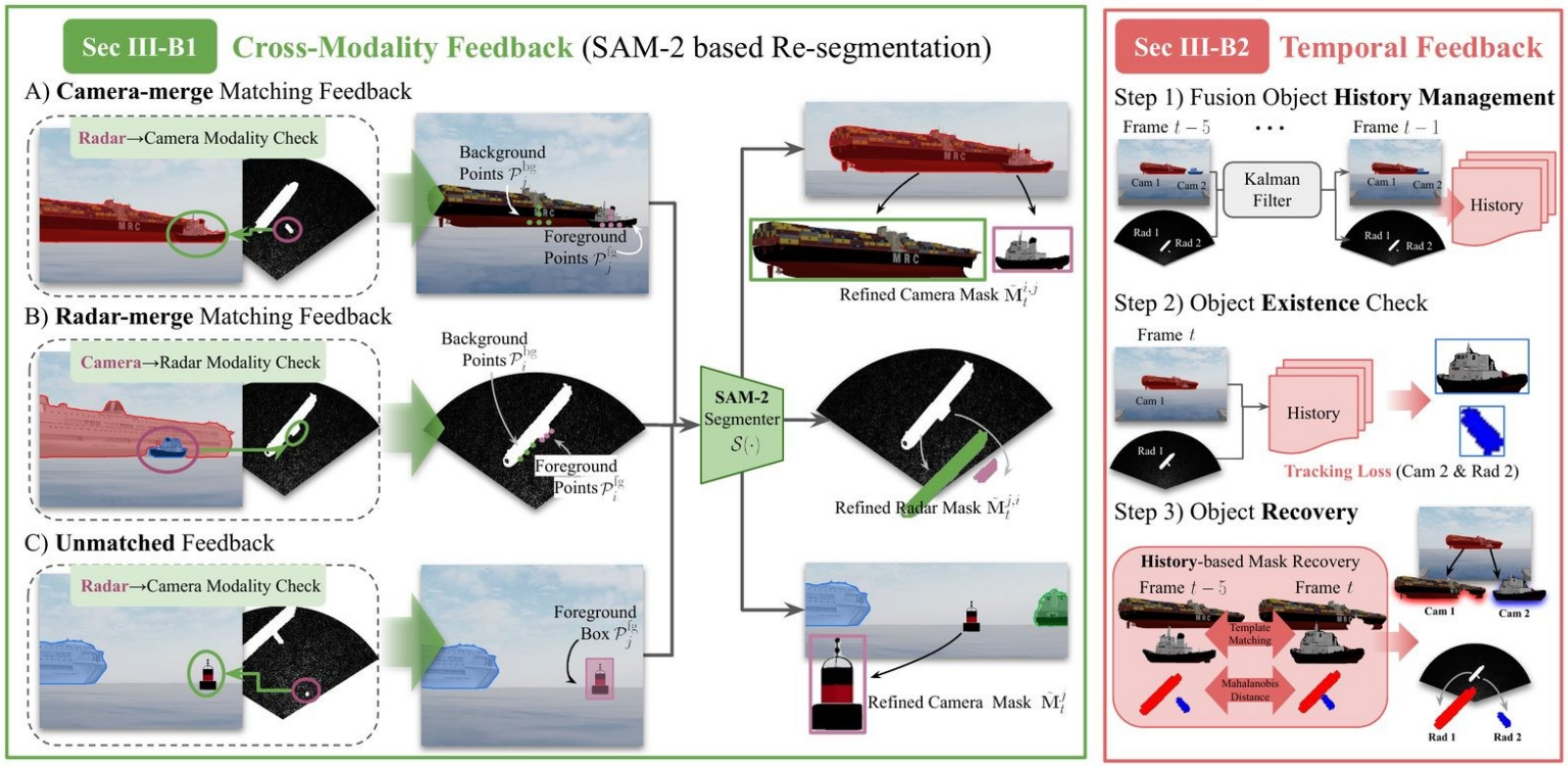
Re-segmentation via cross-modality and temporal feedback

3 CRF-based selection

Energy minimization selects a globally consistent object set

STAGES 2 & 3

Refine with feedback, then select globally



Perception refinement

Cross-modality: other-sensor objects prompt SAM 2 to split or recover masks.

Temporal: Kalman history + template matching restore lost objects.

Object-level CRF

Each fusion hypothesis is a node. Unary and pairwise energies score agreement and conflicts; mean-field + Hungarian picks one hypothesis per object.

RESULTS

Best F1 on all four datasets

Method	Amnam	Pohang	MOANA	Sim.	Overall F1	RadIoU	CamIoU
Bearing	76.73	97.40	70.45	84.54	83.97	0.678	0.484
Distance	80.16	97.40	83.44	87.91	87.86	0.735	0.499
D+V	70.55	96.12	77.61	87.33	86.76	0.720	0.568
AnytimeFusion	87.31	95.97	91.00	86.26	86.91	0.723	0.601
PRFusion (ours)	**95.56**	**100.00**	**96.77**	**99.29**	**99.10**	**0.902**	**0.770**

+19.9 %p

Overall recall vs. Distance
78.67 → 98.54%

+11.2 %p

Overall F1 vs. Distance
87.86 → 99.10%

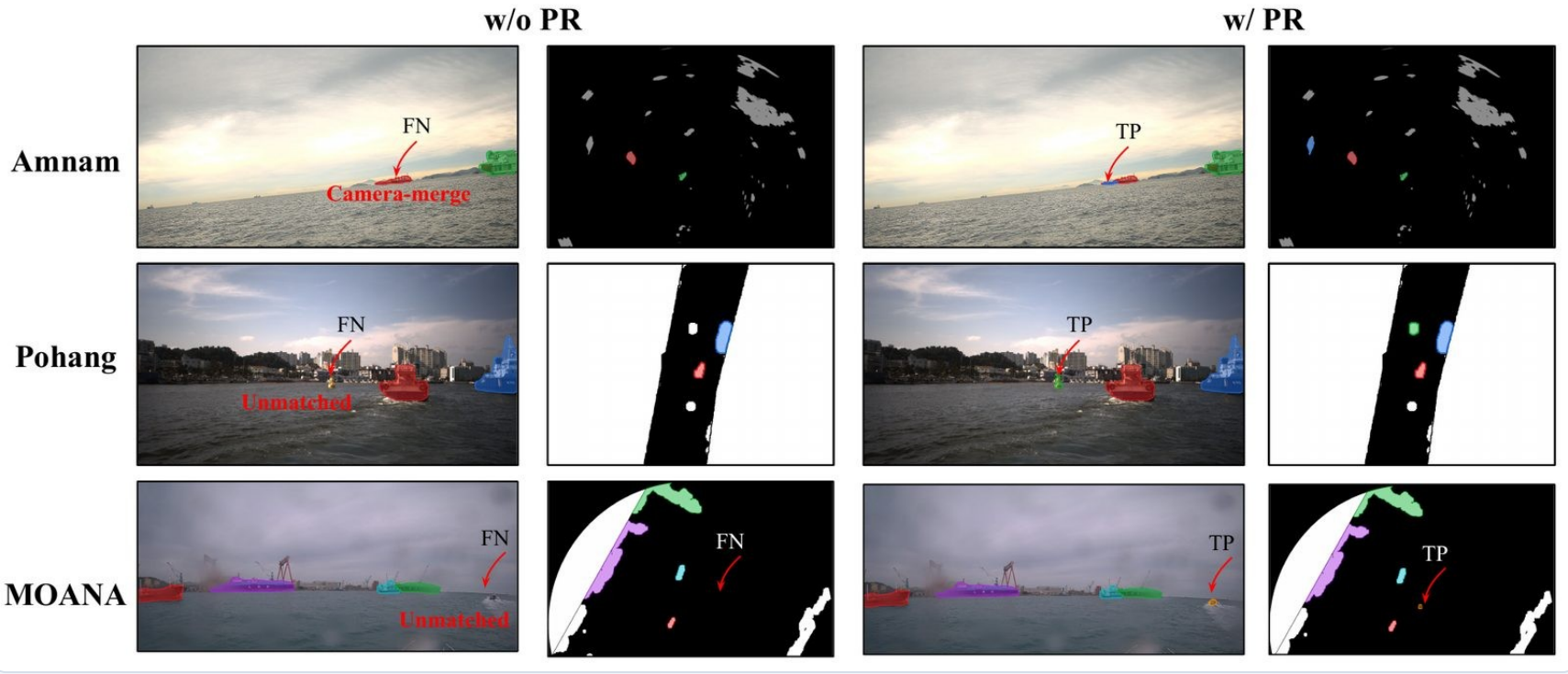
+28.1 %

Camera mask IoU vs. AnytimeFusion
0.601 → 0.770 (relative)

F1 in %. Real-world: Amnam, Pohang Canal, MOANA · Simulation: Gazebo, 30 scenarios · Real-world F1 gain: 2.6–8.3 %p

RESULTS · QUALITATIVE

Refinement recovers missed objects



Amnam

Camera-merge split by cross-modality feedback

Pohang Canal

Radar-only buoy recovered in the camera

MOANA

Lost in both sensors; restored by temporal feedback

Simulation (day / night / fog): all three failure types, including radar-merge, are recovered consistently.

CONCLUSION

Refine, don’t just match

1

Failure as a trigger

Cross-modal and temporal feedback re-segments merged or missed objects at their source.

2

Global consistency

The object-level CRF keeps both precision and recall where NN or Hungarian lose one.

3

Robust at sea

Best F1 on all four datasets, with higher radar and camera mask quality.

Thank you

Benchmark & annotations: sites.google.com/view/prfusion

yeongha.shin@kaist.ac.kr